Com S 4720/5720 Fall 2025

Problem Set 2 (40 pts)

Due at 11:59pm

Friday, September 12

All the problems are from *https://aimacode.github.io/aima-exercises/* (possibly) with minor typo fixes, slight rewriting, or addition of missing information.

1.(13 pts) (Exercise 3.7) The n-queens problem, informally introduced in the textbook, is defined on page 12 of the PPT lecture notes titled "3b. problem solving.pptx", which was posted on September 4. Consider the n-queens problem using an "efficient" incremental formulation given below.

- *States*: All possible arrangements of k queens ($0 \le k \le n$), one per column in the leftmost k columns, with no queen attacking another.
- *Initial state*: No queen on the board.
- *Actions*: Add a queen to any square in the leftmost empty column such that it is not attacked by any other queen.

(a) (10 pts) Explain why the state space has at least $\sqrt[3]{n!}$ states. (*Hint*: Derive a lower bound on the branching factor by considering the maximum number of squares that a queen can attack in a single column.)

(b) (3 pts) Estimate the largest n for which exhaustive exploration is feasible, assuming that i) the most powerful supercomputer can process 200,000 trillion calculations per second, and ii) every state can be explored in one calculation.

2.(12 pts) (Exercise 3.17) Which of the following are true and which are false? Explain your answers.

(a) (3 pts) Depth-first search always expands at least as many nodes as A* search with an admissible heuristic.

(b) (3 pts) $h(n) = 0$ is an admissible heuristic for the 8-puzzle.

(d) (3 pts) Breadth-first search is complete even if zero step costs are allowed.

(e) (3 pts) Assume that a rook can move on a chessboard any number of squares in a straight line, vertically or horizontally, but cannot jump over other pieces. Manhattan distance is an admissible heuristic for the problem of moving the rook from square A to square B in the smallest number of moves.

3.(10 pts) (Exercise 3.18) Consider a state space where the start state is number 1 and each state k has two successors: numbers $2k$ and $2k+1$.

(a) (4 pts) Draw the portion of the state space for states 1 to 15.

(b) (6 pts) Suppose the goal state is 11. List the order in which nodes will be visited for breadth-first search, depth-limited search with limit 3, and iterative deepening search.

4.(5 pts) (Exercise 3.22) Describe a state space in which iterative deepening search performs much worse than depth-first search (for example, $O(n^2)$ vs. $O(n)$).